

HVIA x OLIST

H V I A A D V I S O R Y

Brazilian E-Commerce Business Analysis & AI Solution Proposal

An evidence-based review of delivery performance in the Olist public e-commerce dataset, and a proposed predictive delivery-risk solution for Olist's Data/AI and Logistics teams.

Executive Summary

This business case analyzes the publicly available Olist Brazilian E-Commerce dataset (approximately 100,000 orders, 2016–2018) to identify a logistics-related business opportunity and to propose an AI/ML solution that HVIA could deliver to Olist's Data/AI and Logistics teams.

PROBLEM

Delivery delays and freight costs vary meaningfully across orders and customer states, and this variation is not fully visible until after an order has already been affected.

KEY EVIDENCE

An overall observed delay rate of 6.44%; delay rates ranging from 13.08% to 20.09% across the five states analyzed; and a review-score gap of 4.15 (on-time) vs. 2.26 (delayed).

BUSINESS OPPORTUNITY

Earlier visibility into orders that are more likely to experience a delivery delay, so that Logistics and Operations teams can investigate and intervene before the delay reaches the customer.

PROPOSED SOLUTION

A Predictive Delivery Risk Engine that scores each order's delay probability and surfaces the contributing risk factors to the Data/AI team, for onward use by Logistics and Operations.

FRAMING

Using the public Olist dataset, we identified a measurable logistics pattern. This suggests an opportunity for earlier delivery-risk visibility. HVIA could address this through a predictive delivery-risk engine that turns operational data into actionable signals for the Data/AI and Logistics teams.

SCOPE NOTE

All findings below come from analysis of the public Olist dataset only. This document does not claim to know Olist's internal systems, financial impact, or the causes of the observed patterns — it presents associations observed in the data and a proposed direction for further investigation.

Company Research — Olist

What Olist Does

Olist is a Brazilian technology company, founded in 2015 and headquartered in Curitiba, Paraná, that provides an ecosystem of software solutions for retailers and small and medium-sized businesses selling online. Its core offering today is an ERP system — built around its Tiny acquisition — that unifies inventory, order management, invoicing, and financial tracking for sellers, while also connecting those sellers to major marketplaces such as Amazon, Mercado Livre, Shopee, and Magalu, as well as logistics and payment partners.

Why Logistics Is Relevant

Because Olist's ecosystem sits between sellers, marketplaces, customers, and shipping/logistics partners, order fulfillment and delivery performance are a natural part of the value it manages for the businesses it serves. The public Olist e-commerce dataset reflects this: it contains order-level delivery timestamps, estimated delivery dates, freight values, customer locations, and post-purchase review scores — all of which relate directly to logistics and delivery experience.

NOTE

This section describes Olist's publicly stated business model. It does not claim any specific internal logistics arrangement, carrier relationship, or operational capability beyond what is publicly available.

Dataset Understanding

Dataset Overview

The analysis uses the Olist Brazilian E-Commerce Public Dataset, which contains approximately 100,000 orders placed between 2016 and 2018, covering orders, order status, purchase and delivery timestamps, estimated delivery dates, products, sellers, customers and customer location, freight value, product price, product weight, reviews, and payments.

Relevant Data Fields

- order_id, customer_id, order_status
- order_purchase_timestamp, order_delivered_customer_date, order_estimated_delivery_date
- delivery_time_days, delay_days, is_delayed
- seller_id, price, freight_value
- review_score
- customer_state, customer_city
- product_category_name, product_weight_g

Business Problem Identified

The analysis found meaningful variation in delivery performance and freight costs across orders and customer states. This creates a potential opportunity to improve visibility into the factors associated with delivery delays and high shipping costs.

IMPORTANT FRAMING

This analysis does not indicate that Olist is losing money, or that Olist has a confirmed logistics crisis, or that any internal Olist logistics system is failing. It identifies a potential logistics visibility opportunity based on patterns observed in the public dataset.

Business Analysis & Key Findings

Overall Delay Rate

Across all analyzed orders, the overall observed order delay rate was **6.44%**.

Regional Analysis

Delivery delay rates vary significantly across customer states in the analyzed dataset. The table below shows the five states with the highest observed delay rates.

State	Total Orders	Delay Rate	Avg Freight Value	Avg Delivery Days
AL	411	20.09%	R\$ 35.74	23.98
MA	740	17.37%	R\$ 38.20	21.14
SE	345	15.84%	R\$ 36.65	20.98
CE	1,327	13.15%	R\$ 32.71	20.53
PI	493	13.08%	R\$ 39.11	18.93

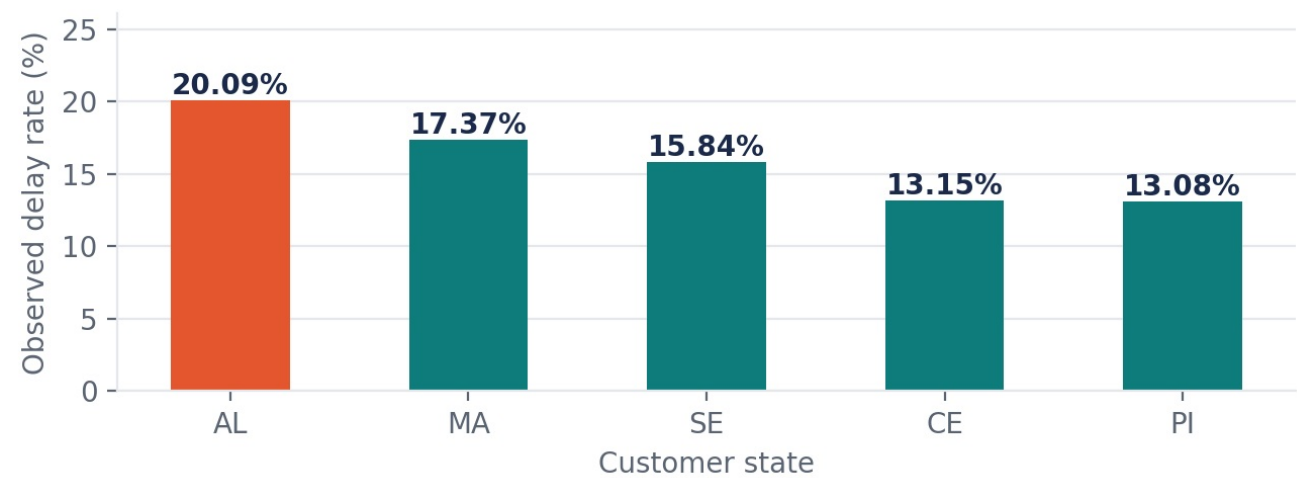


Figure 1 — Observed delay rate by customer state (five states analyzed)

AL had the highest observed delay rate in this state-level output, at 20.09%. Freight value alone does not fully explain this variation: PI has a higher average freight value (R\$ 39.11) than AL (R\$ 35.74), yet a lower delay rate (13.08% vs. 20.09%) and a shorter average delivery time (18.93 vs. 23.98 days). This suggests that **freight value is an important observed signal, but it does not fully explain delivery performance** — other factors captured in the dataset likely also play a role.

Freight Analysis

Orders were divided into three freight-value groups (Low, Medium, High) using quantile-based binning. Higher freight-value groups were associated with higher observed delay rates and longer average delivery times.

Freight Group	Total Orders	Delay Rate	Avg Delivery Days
Low Freight	32,689	4.34%	7.89
Medium Freight	33,459	7.01%	13.17
High Freight	33,869	7.96%	15.00

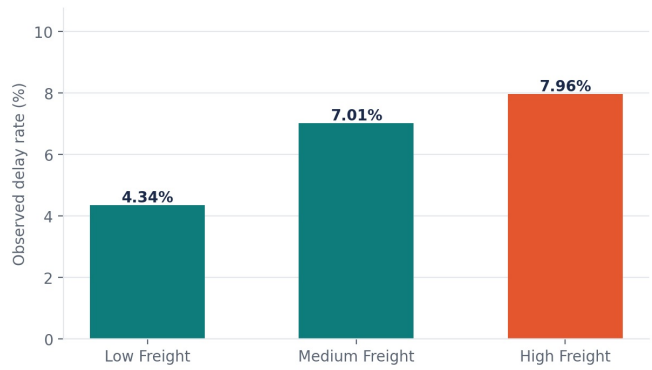


Figure 2 — Delay rate by freight-value group

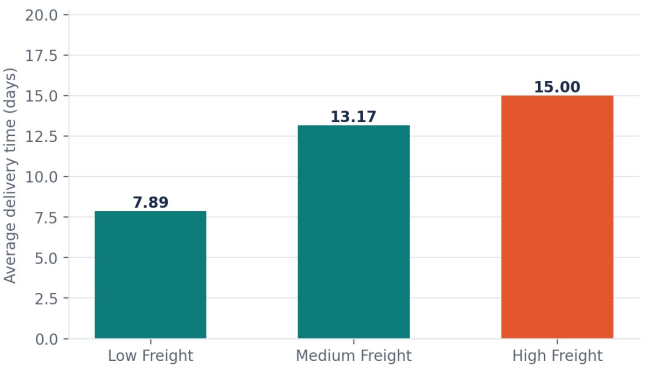


Figure 3 — Average delivery days by freight-value group

IMPORTANT FRAMING

The analysis does not show that higher freight causes delays. It shows that higher freight-value groups were associated with higher observed delay rates and longer delivery times in the analyzed dataset.

Customer Experience & Business Story

Customer Review Analysis

Delayed orders were associated with substantially lower average review scores than on-time orders.



Figure 4 — Average review score, on-time vs. delayed orders

Order Type	Avg Review Score
On-time orders	4.15 / 5
Delayed orders	2.26 / 5

Key Business Insights

- **Regional variation:** Delivery delay rates vary significantly across customer states, with a roughly 7-point spread across the five states analyzed (13.08%–20.09%).
- **Freight / delivery relationship:** Higher freight-value groups show higher observed delay rates and longer average delivery times — though freight alone does not fully explain performance (e.g., PI vs. AL).
- **Customer experience relationship:** Delayed orders are associated with substantially lower review scores (2.26 vs. 4.15).

BUSINESS NARRATIVE

Delivery performance varies across customer states. Higher freight-value groups show higher observed delay rates and longer average delivery times. Delayed orders are associated with substantially lower review scores. Taken together, this suggests a potential opportunity: Olist could benefit from better early visibility into orders that are more likely to experience delivery delays.

KEY BUSINESS QUESTION

Can potentially high-risk orders be identified before the delay happens, so the relevant teams can investigate and take preventive action?

Proposed HVIA Solution

Predictive Delivery Risk Engine

HVIA proposes developing a machine-learning system that estimates the probability that a given order will be delayed, using available order, geographic, product, and freight-related characteristics together with other relevant historical features. For each order, the system would produce an output such as:

Order #123 — Delay Risk = 78%

The system is designed to go beyond a single risk score by also surfacing the contributing risk factors behind each prediction, for example:

- High freight value
- Destination region with elevated historical delay rate
- Long expected delivery time
- Other relevant model-supported signals

How It Works



Business Value

The value of this solution is not simply having an AI model — it is changing the operating workflow from *discovering the problem after the order is delayed*, to *identifying potentially high-risk orders early enough for teams to investigate and take preventive action*.

Future Extension — Carrier Optimization

If carrier-level operational data becomes available, the risk engine could in the future be extended to incorporate carrier performance and support data-driven carrier selection or routing decisions. The current public dataset does not provide enough evidence to implement carrier optimization today; it is presented here only as a possible future direction requiring additional carrier-level data.

Outreach Strategy & Message

| Target Stakeholder: Data / AI

The recommended point of contact is a Data / AI professional at Olist, rather than a Logistics or Operations contact directly.

Why This Stakeholder

The proposed solution is fundamentally an ML prediction system. A Data/AI professional is best positioned to evaluate the model's feasibility, data requirements, and integration into Olist's existing data ecosystem before its outputs are ever operationalized.

How the Output Reaches Logistics / Operations

Once validated by the Data/AI team, the risk scores and risk-factor explanations are designed to be passed downstream as actionable signals that Logistics and Operations teams can use to investigate and intervene on high-risk orders before delays occur.

| Outreach Message

Hi [Name],

I recently analyzed the public Olist e-commerce dataset and found an interesting logistics pattern: higher-freight orders were associated with longer delivery times and higher observed delay rates, while delayed orders had substantially lower average review scores (2.26 vs. 4.15 for on-time orders).

This made me think there may be an opportunity to use predictive ML to identify high-risk orders before delays happen, along with the factors contributing to that risk, so the Data/AI team could provide actionable signals to Logistics and Operations.

I'd be interested in discussing whether this kind of predictive delivery-risk workflow could fit into Olist's existing data ecosystem.

— HVIA

Final Business Case

The complete case can be summarized as a single chain of reasoning, from observed evidence to a concrete next step:



CLOSING STATEMENT

Using the public Olist dataset, we identified a measurable logistics pattern. This suggests an opportunity for earlier delivery-risk visibility. HVIA could address this through a predictive delivery-risk engine that turns operational data into actionable signals for the Data/AI and Logistics teams.

All numerical findings in this document originate from the analysis of the Olist Brazilian E-Commerce Public Dataset conducted for this business case. Company information about Olist is drawn from Olist's publicly available company materials and product/company profiles. This document presents associations observed in the data and proposed opportunities; it does not assert confirmed internal problems, causal relationships, or unverified claims about Olist's operations.